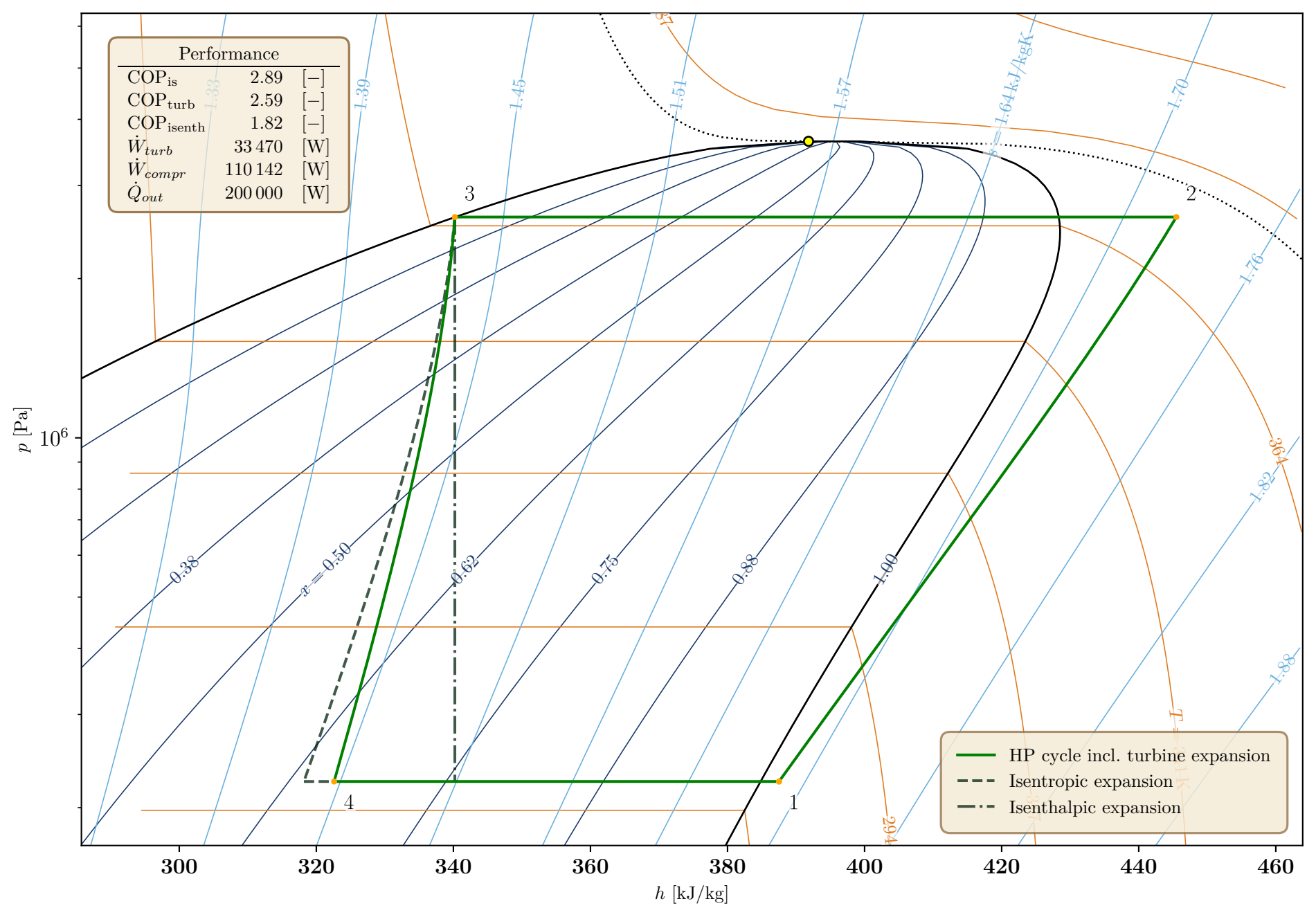

Performance
COP_is 2.89 [−]
COP_turb 2.59 [−]
COP_isenth 1.82 [−]
Ẇ_turb 33 470 [W]
Ẇ_compr 110 142 [W]
Q̇_out 200 000 [W]
3
2
4
1
HP cycle incl. turbine expansion
Isentropic expansion
Isenthalpic expansion
p [Pa]
10^6
h [kJ/kg]
300
320
340
360
380
400
420
440
460
x = 0.50
0.38
0.62
0.75
0.88
1.00
s = 1.644 kJ/kgK
1.33
1.39
1.45
1.51
1.57
1.70
1.76
1.82
1.88
364
294